

i3 Technologies — Full Platform Implementation Planning Document

Unified Project: OTT Platform · AI Lab · EvalOS · Admissions AI · 8 Industry Solutions

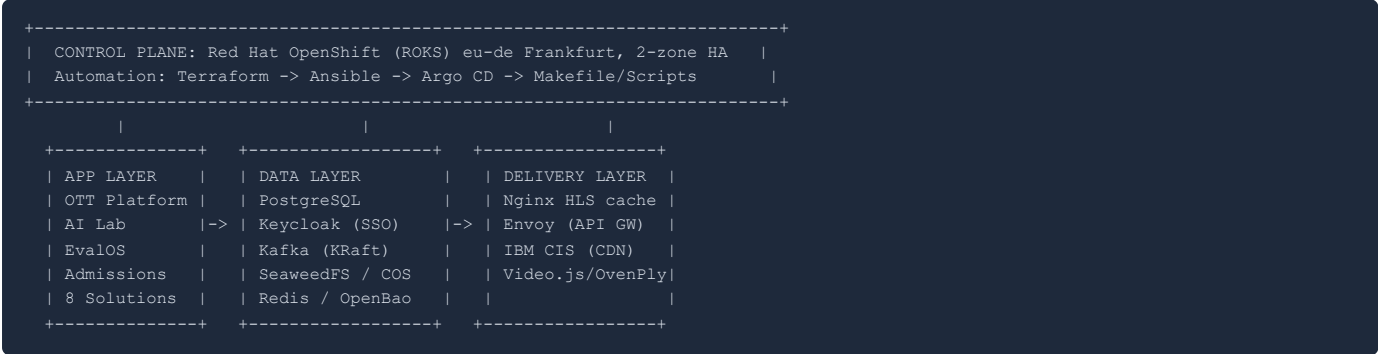
Version 1.0 | July 2026 | Single ROKS Cluster | eu-de Frankfurt | Synthesised from 17 planning assets

1. Executive Summary

TOTAL IBM CLOUD CREDIT \$26,467.77 4 pools · 21-month runway	MONTHLY CAP ≤\$1,500/mo Steady-state ~\$1,250 · ~\$250 buffer	PLATFORM COMPONENTS 5 Major + 8 One ROKS cluster · 17 namespaces
TARGET GO-LIVE 20 Jul 2026 First student cohort onboarded	STUDENT SCALE 5,000–10,000 2,000 peak concurrent	COUNTRIES SERVED 7 African Kenya, Ghana + 5 others

Strategic Intent: All five platform components run on *one* ROKS cluster sharing PostgreSQL, Keycloak, Kafka, vLLM, and observability — eliminating infrastructure duplication. Adding OTT + AI Lab + EvalOS + Admissions to the existing 8-solution cluster costs **marginal ≈\$0** in new base infrastructure.

2. Platform Architecture Overview



17-Namespace Cluster Topology

vault – OpenBao secrets	shared-infra – PostgreSQL, Kafka, Redis, Prometheus
security – Keycloak, Falco, Zeek	solution-01...08 – 8 industry AI solutions
ai-lab-platform [NEW] – RHOAI, vLLM	rhoai-platform [NEW] – InstructLab, Model Registry
admissions-assistant [NEW] – Chatbot, ChromaDB	evalos-platform [NEW] – Assessment, Sandbox
ai-lab-cohort-{n} [NEW] – Per-cohort workbenches	ott-platform [NEW] – OME, Directus, SeaweedFS, Nginx

Shared Services (Cross-Namespace)

Service	Namespace	Consumers	Purpose
PostgreSQL (Crunchy Operator)	shared-infra	All components	Primary database, operator-managed HA
Keycloak	security	All	SSO, OIDC/SAML, enterprise IdP federation
OpenBao	vault	All	Secrets management, zero hardcoded credentials
Kafka (Strimzi 3.7 KRaft)	shared-infra	Event-driven workflows	Message broker; ZooKeeper-free
vLLM (Granite 3.1 2B)	ai-lab-platform	AI Lab, EvalOS, Admissions	Shared LLM inference; KEDA HPA-scaled
Grafana + Prometheus	shared-infra	All	Metrics, dashboards, cost monitoring
Falco + Zeek	security	All	Runtime security + network IDS

3. The 5 Major Platform Components

1

OTT Live Streaming Platform

+\$150–250/mo additive

Self-hosted live delivery of bootcamps, AI HALISI broadcasts, and client training with VOD replay across 7 African countries. **MinIO NOT used** — archived read-only 25 Apr 2026. SeaweedFS (Apache 2.0) is the replacement.

```
Instructor (OBS Studio RTMP/SRT) -> OvenMediaEngine (LL-HLS + WebRTC out)
-> Nginx HLS segment cache + Envoy API Gateway
-> Video.js (VOD) / OvenPlayer (live, WebRTC+LL-HLS fallback)
-> Learners in 7 countries [240p/300kbps floor -> 1080p ABR]
Storage: SeaweedFS primary (S3-compatible) + IBM COS DR/cold archive
CMS:      Directus + Keycloak + PostgreSQL + n8n automation
```

Layer	Tool	License	Key Reason
Live ingest/transcode	OvenMediaEngine	AGPLv3	Sub-second LL-HLS; SRT packet-loss recovery for mobile hotspots
VOD batch transcode	FFmpeg	LGPL/GPL	Full ABR ladder control; 240p floor for African mobile data
Facilitator encoder	OBS Studio	GPLv2	Free, SRT-native, zero infra cost for facilitators
Headless CMS	Directus	GPLv3/BSL	Database-first; non-technical admin UI; REST + GraphQL APIs
Auth / SSO	Keycloak	Apache 2.0	Red Hat-backed; OIDC/SAML; enterprise IdP federation-ready
Workflow automation	n8n	Fair-code	Class end → archive → Directus update → cohort notify
Live viewer player	OvenPlayer	MIT	Native LL-HLS + WebRTC with automatic fallback
VOD replay player	Video.js + VHS	Apache 2.0	Largest plugin ecosystem; works on older Android devices
Primary object store	SeaweedFS	Apache 2.0	MinIO successor; HLS segment-efficient; S3-compatible
DR / cold archive	IBM Cloud COS	Commercial	Credit-funded; \$10/TB/mo; S3 replication target from SeaweedFS
Media segment cache	Nginx	BSD-2	Industry-standard HTTP caching for .ts/.m4s HLS segments
API gateway	Envoy	Apache 2.0	OpenShift-native; circuit breaking; retry logic
VOD catalog (optional)	PeerTube	AGPLv3	P2P-assisted playback reduces egress; disable federation

ABR Encoding Ladder (Low-Bandwidth Priority)

Rendition	Resolution	Bitrate	Target Audience
Floor (auto-quality default)	426x240	300 kbps	Kenya/Ghana mobile data
Low	640x360	600 kbps	Mid-range mobile data
Mid	854x480	1,000 kbps	Good mobile / WiFi
High	1280x720	2,500 kbps	Office broadband
Full HD	1920x1080	5,000 kbps	Studio / data-centre viewers

2 AI Lab Platform (5,000–10,000 Students)

Phase 1 ~\$0 / Phase 2 ~\$400–600/mo

Critical distinction: Free APIs work for each student’s personal workflow. They DO NOT work as a shared platform (1,000–2,000 concurrent requests = limits exhausted in seconds). Phase 1 = personal per-student keys only. Phase 2 = shared vLLM on ROKS.

Phase 1 — Per-Student Free API Stack (Weeks 1–12)

Provider	Daily Free Limit	Use Case
Groq	14,400 req/day	Real-time inference, speed-critical tasks
Cerebras	1M tokens/day	Batch processing, large context jobs
OpenRouter	200 req/day (29 free models)	Model diversity, experimentation
Google AI Studio	1,500 req/day (Gemini 3 Flash, 2M context)	Multimodal, document understanding
Mistral	1B tokens/month	EU residency, code tasks
Ollama (local)	Unlimited (hardware-bound)	Privacy, offline capability

Phase 2 — 10 SLMs Mapped to Lab Modules (Month 4+)

Module	Model	Deployment	SWE-bench Tier
General Reasoning	Llama 3.3 70B	Groq API (personal)	A (30–40%)
Coding Track	DeepSeek Coder V2 Lite	Local/Ollama (~18 GB)	A+ specialised
Lightweight Tasks	Phi-4 14B	Local/Ollama (~9 GB)	A efficient
Fast Inference	Llama 3.1 8B	Groq / ~5 GB local	B utility
Industry Fine-Tuning	IBM Granite 3.1 2B	vLLM on ROKS (shared)	A+ tunable
SLM Scale / EU	Mistral Small 3.1	vLLM on ROKS (24B)	A+ EU-compliant
Multimodal	Gemini 3 Flash	Google AI Studio API	A+ multimodal
A+ Reasoning	DeepSeek V4	OpenRouter (cache \$0.03/1M tokens)	A+ cost-efficient
High Volume Batch	Qwen3.6-Plus	Cerebras API	S (65–79% SWE)

3 EvalOS — Student Assessment Engine

~\$80/mo marginal

Website -> AI Chat (2-10 min) -> Registration (2 min) -> Pre-screen (15 min)
-> Resume AI (30s) -> Cognitive (25 min) -> Coding (60 min)
-> AI Interview (20 min, Granite 3.1 2B) -> Auto-Enrolment (60s) -> AI Lab Access

4 Track-Specific Execution Profiles

Profile	Track	Runtime	Grading
A — Python/AI	AI Engineering	Ubuntu + Python 3.11 + PyTorch + HuggingFace	Hidden test cases; stdout comparison; time/memory limits
B — IaC	Cloud/DevOps	Terraform + kubectl + AWS/Azure CLIs	plan/dry-run; state JSON parse; resource spec validation
C — Robotics/Rust	Robotics/Edge AI	Rust/C++ + ROS 2 + Gazebo simulator	Compile → run → monitor ROS topics; data-structure check
D — Cyber/Forensics	Security	Target MicroVM (vulnerable) + Student MicroVM (Kali)	Grade by target state changes — not student code analysis

Security & Anti-Cheat Controls

Control	Implementation	Purpose
Zero-Trust Execution	Firecracker MicroVM per submission (~125 ms boot)	Prevent sandbox escape / container breakout
Network Blackholing	Zero internet access (whitelisted APIs Week 1 only)	Prevent cheating via web search
Resource Limits	CPU/memory cgroups; kill at 5 s timeout	Prevent DoS from infinite-loop code
Clipboard Lockdown	Ctrl+C/V intercepted; clipboard events disabled	Prevent copy-paste cheating
Keystroke Logging	xterm.js capture in live terminal sessions	Validate system navigation skills
Plagiarism: AST + MOSS	Abstract Syntax Tree comparison + Stanford MOSS (auto-trigger)	Catch logic-similar code despite variable renaming
Partial Credit	3/5 test cases pass = partial marks	Fair grading of near-complete solutions

4 Admissions AI Assistant

~\$30/mo · Fastest-to-Ship: Day 10

Component	Tool	Notes
Primary LLM	IBM Granite 3.1 2B	Shared vLLM endpoint (no separate inference cost)
Embeddings	BGE-Large (BAAI)	1 GB VRAM; high-accuracy local retrieval
Vector Database	ChromaDB	Programme info, FAQs, curriculum, requirements
RAG Framework	LlamaIndex	Intelligent chunking + retrieval pipeline
Backend API	FastAPI + WebSocket	Real-time streaming chat responses
Safety Testing	Promptfoo	Red-teaming for prompt injection vulnerabilities
Automation	n8n (shared)	Lead → EvalOS trigger → AI Lab provisioning chain
CDN Delivery	IBM Cloud CIS	Widget asset delivery + WAF protection
Quality Monitoring	RAGAS	Faithfulness + Relevancy scores; alert on degradation

5 8 Industry AI Solutions

Already deployed · Marginal \$0

All 8 solutions continue running undisturbed in their existing namespaces, sharing Keycloak SSO, OpenBao secrets, Kafka messaging, Grafana/Prometheus observability, and Falco/Zeek security. Adding the four new components to the cluster costs **marginal ≈\$0** in new base infrastructure.

4. IBM Cloud Credit Master Plan

26PENEW18K (PRIMARY) \$18,000 Phase A: Jul 2026 – Aug 2027	BUYBRS (RESERVE) \$5,000 Phase B: Sep 2027 – Oct 2028	POWERVS2500 (GPU-ONLY) \$2,467.77 InstructLab fine-tuning only
---	--	---

Steady-State Monthly Breakdown (Month 7+)

Service	Cost Range	Notes
ROKS worker nodes	\$290	2–4x bx2.4x16; autoscale 2–4 nodes
vLLM inference (shared)	\$100–200	Granite 3.1 2B + Mistral Small serving all 3 new components
OTT additive (OME + SeaweedFS + Nginx)	\$150–250	cx2 pods for OME; SeaweedFS PVCs; Nginx cache nodes
IBM Cloud CIS (CDN/WAF)	\$40–60	Admissions widget + OTT delivery
IBM Cloud COS	\$30–80	OTT DR archive + EvalOS code storage (~\$10/TB/mo)
EvalOS PVC + Sandbox burst	\$35–105	PostgreSQL PVCs + Firecracker burst during exam windows
Secrets Manager + DNS	\$15–20	OpenBao supplement + VPC networking
GPU training (PowerVS pool)	\$35–175	InstructLab fine-tuning; separate pool; provision → train → destroy
TOTAL	\$695–1,180	Within \$1,500/mo cap · \$320–805 monthly buffer

5. 4-Layer Automation Stack

```
Layer 1: Terraform (IBM Cloud Provider)
  VPC + ROKS + COS + IAM | Remote state backend on COS [CRITICAL FIX from code review]
  |
  v [OpenBao Manual Gate -- human approves secrets injection between Terraform + Ansible]
  |
Layer 2: Ansible (12 Roles: 8 existing + ai-lab + evalos + admissions + ott-platform)
  Configuration management | Operator bootstrap | Initial service setup
  |
  v [Git push triggers GitOps sync]
  |
Layer 3: Argo CD (15 Applications: 11 existing + ai-lab + evalos + admissions + ott-platform)
  GitOps continuous sync | Automated pruning + self-healing (non-vault apps)
  |
  v
Layer 4: Makefile  make deploy | make gpu-up | make gpu-down | make verify
```

Code Review Critical Fixes (Must Resolve Before Production)

Issue	Severity	Fix	Effort
Terraform local .tfstate — no remote backend	CRITICAL	Add COS backend config to ALL Terraform modules	30 min
ApplicationSet generator too permissive	HIGH	Restrict generators; add PR approval gate	1 hour
Kafka on ZooKeeper (Strimzi 3.7 dropped ZK)	HIGH	Migrate to KRaft mode	1 hour
Hardcoded gpt-4-turbo-preview in inference code	MEDIUM	Replace with configurable Granite 3.1 2B endpoint variable	20 min
Docker-in-Docker in EvalOS sandbox spec	MEDIUM	Standardise on Firecracker/gVisor — no DinD in production	1 sprint

6. Master Implementation Timeline (6-Month Roadmap)

PRE-BUILD Days 1–3 ~3–4 hrs	- Verify \$26,467.77 credit across all 4 pools; confirm Red Hat entitlements (OCP, RHOAI, InstructLab, vLLM, Satellite 50-node, Partner 500-node) - Install CLI toolchain in WSL2 Ubuntu: Terraform ≥ 1.6, oc, ibmcloud CLI + plugins, Helm ≥ 3.15, Python 3.12+, Ansible, Make, ilab, Ollama, Git 2.43+ - Collect all 7 free API keys: Groq, Cerebras, OpenRouter, Google AI Studio, Mistral, HuggingFace, MOSS <i>Laptop (Win 10, i5-6200U, 7.4 GB RAM): orchestration only via WSL2 — all compute on ROKS</i>
MONTH 1 Days 1–30 Admissions AI Live	- Day 1: Verify cluster state; confirm all credit pools active - Days 1–2: Terraform VPC + ROKS + COS (conditional if cluster exists, 25–45 min) - Days 2–3: Ansible bootstrap — OpenBao manual gate + 9 existing roles (45–90 min) - Days 3–7: Build Admissions AI (FastAPI + ChromaDB + BGE-Large + n8n workflows) Days 7–10: DNS + IBM CIS CDN widget deploy → Chatbot LIVE on website - Days 10–30: EvalOS DB schema, Keycloak auth, namespace scaffolding, Firecracker runtime
MONTH 2 Days 31–60 EvalOS + OTT Pilot	- EvalOS: Firecracker/gVisor runtime; FastAPI backend + Redis queue; Monaco Editor shell - OTT: Deploy OvenMediaEngine Origin pod → end-to-end live class test (1 class streamed) - OTT: SeaweedFS single-node + IBM COS DR replication; Directus CMS + Keycloak; Nginx HLS cache - AI Lab: Per-student API setup guide published; Open WebUI + ChromaDB self-hosted (~50 concurrent)
MONTH 3 Days 61–90 Profiles A+B + OTT Prod	- EvalOS Profile A (Python/AI) full sandbox pipeline — Python/AI assessments live - EvalOS Profile B (IaC) Terraform dry-run parser — IaC track assessable - OTT: OME Origin/Edge separation; Envoy API GW; Video.js + OvenPlayer portal; SeaweedFS multi-volume HA - AI Lab: RHOAI operator installed; vLLM endpoint (Granite 3.1 2B) live — shared inference for all components
MONTH 4 Days 91–120 All Profiles + AI Lab Prod	- EvalOS Profile C (Rust/ROS2/Gazebo) + Profile D (Cyber dual-MicroVM) — all 4 tracks live EvalOS AI Interview (Granite 3.1 2B) — full 5-stage pipeline complete end-to-end - AI Lab: GPU pool initialised (gx2.8x64 V100, \$0 at idle); InstructLab fine-tuning validated - AI Lab: Per-cohort RBAC; RHOAI workbench auto-provisioning on enrolment within 60 s - OTT: Full UDP firewall audit (UDP 10000–10004); PeerTube VOD catalog decision
MONTH 5 Days 121–150 Dashboard + VOD + AutoEnrol	- EvalOS: Instructor dashboard (question creation, test case editor, rubric config); student gradebook + analytics - EvalOS: Plagiarism pipeline (AST + MOSS auto-triggered pre-review) - OTT: FFmpeg VOD batch pipeline; n8n post-class chain; full React learner portal (schedule, catalog, live, replay) Admissions: Full auto-enrolment — EvalOS PASS → AI Lab namespace provisioned in 60 seconds
MONTH 6 Days 151–180 Load Test + Go-Live	- EvalOS: 100 concurrent submission storm; instructor mock exam; UX debug and grading accuracy - OTT: Multi-country stream test (7 countries); 240p/300 kbps floor validated; SRT packet-loss resilience - AI Lab: 2,000 concurrent users; vLLM HPA + KEDA scale test; GPU burst automation (make gpu-up/down) - Platform: OpenBao rotation test; Argo CD drift test; Falco/Zeek 24-hour audit; Grafana completeness review TARGET GO-LIVE: 20 July 2026 — First student cohort onboarded across all 5 platform systems

7. Risk Register

Critical / High Risks

#	Risk	Component	Severity	Mitigation
R1	CPU exhaustion during concurrent live classes	OTT	Critical	Dedicated cx2 node pool for OME; Origin/Edge separation; GPU NVENC at >4-6 concurrent channels
R2	UDP ports blocked - SRT/WebRTC fails at go-live	OTT	Critical	Verify VPC security groups + NetworkPolicies allow UDP 10000-10004 BEFORE go-live
R3	Sandbox escape via MicroVM vulnerability	EvalOS	Critical	Firecracker (not Docker); gVisor; zero network; instant destroy after every submission
R4	Free API rate limits at shared-platform scale	AI Lab	High	Phase 1 = personal per-student keys ONLY; Phase 2 = shared vLLM on ROKS
R5	Terraform local state corruption / loss	Platform	High	Remote state backend on COS (critical fix); state locking; periodic backup

Medium Risks

#	Risk	Component	Mitigation
R6	RAG hallucination in Admissions chatbot	Admissions	RAGAS Faithfulness; Promptfoo red-team; human review fallback
R7	Egress cost creep from VOD replay	OTT	Nginx caching; PeerTube P2P; IBM Billing alert at 200GB/mo
R8	DB contention at live class start time	OTT	Isolate PostgreSQL/Keycloak node pool from OME transcoding
R9	Argo CD ApplicationSet drift	Platform	Restrict generators; mandatory PR checks; weekly drift reports
R10	GPU pool left running - cost overrun	AI Lab	make gpu-up / make gpu-down enforced; IBM Billing alert at \$150/day
R11	IBM Cloud credit overrun >\$1,500/mo	All	Grafana cost dashboard; hard alerts at \$1,200 and \$1,500
R12	OpenBao unseal failure	Platform	Test quarterly; backup unseal keys in 2 offline locations
R13	MinIO in existing codebase	Platform	Full grep for minio in config/YAML/env; replace all with SeaweedFS/COS

8. Portability and Exit Strategy

IBM Cloud credit is time-boxed (~21 months). Zero architectural lock-in. Estimated post-credit migration: **1-2 engineering sprints, zero application code changes.**

Component	Portability	Post-Credit Migration
ROKS OpenShift	Portable	Same OCP YAML targets self-managed OCP on any cloud or bare-metal
SeaweedFS	Portable	Apache 2.0; any K8s; S3-compatible; migrates via replication
Keycloak + PostgreSQL	Portable	Standard SQL; pg_dump/restore; Crunchy Operator on any OCP
OpenBao + vLLM	Portable	MPL 2.0 / Apache 2.0; any GPU-equipped K8s cluster
IBM Cloud COS	Switchable	Flip SeaweedFS replication to self-hosted secondary; S3 API compatible
IBM Cloud CIS	Switchable	Replace with Cloudflare free tier or self-hosted Nginx edge; DNS cutover only

9. 6-Month Milestone Summary

Milestone	Target Date	Deliverable
M0 - Pre-Build Complete	Day 3	All tools, credentials, entitlements verified
M1.1 - Admissions AI Live	Day 10	Chatbot on website; lead capture active
M1.2 - EvalOS Foundation	End July 2026	DB schema, auth, namespaces, Firecracker runtime
M2.1 - OTT Pilot Stream	End July 2026	One live class streamed end-to-end on ROKS

M3.1 - EvalOS Profiles A+B	End Aug 2026	Python/AI and IaC tracks fully assessable
M3.3 - vLLM Endpoint Live	End Aug 2026	Shared Granite 3.1 2B serving all components
M4.1 - EvalOS All Profiles	End Sep 2026	All 4 tracks (Python, IaC, Rust/ROS2, Cyber) live
M4.2 - AI Lab Production	End Sep 2026	RHOAI workbenches + GPU burst pool ready
M5.1 - Instructor Dashboard	End Oct 2026	Self-serve question creation, rubrics, gradebook
M5.2 - Auto-Enrolment	End Oct 2026	EvalOS PASS to AI Lab namespace in 60 seconds
M6.1-M6.3 - Load Tests Pass	End Nov 2026	100 concurrent EvalOS; 2,000 AI Lab; 7-country OTT
FULL GO-LIVE	20 July 2026	First student cohort onboarded across all 5 platform systems

Synthesised from 17 i3 Technologies planning assets including: OTT Architecture Blueprint, Unified Platform Replan, EvalOS + Admissions Plans, IBM Cloud Credit Master Plan, Pre-Build Checklist, Automation Code Review, Workspace Alignment Review, AI Lab Master Plan, Architecture Implementation Spec (RAG), Building a Proprietary Assessment Engine, Strategic Infrastructure Roadmap 2026, Model Selection Handbook, and From Tokens to Transformation.

Made with IBM Bob